

Project Initialization and Planning Phase

Date	15 August 2026
Team ID	XXXXXX
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Scenario 1: Small-Scale Farmer

Rajesh is a progressive small-scale farmer based in Uttar Pradesh aiming to maximize crop yield and farm revenue. Unpredictable monsoon shifts and volatile market demands make seasonal crop planning challenging. Through historical data insights (1997–2021), Rajesh seeks seasonal trends, soil-appropriate high-performing varieties, and regional productivity benchmarks to make informed, risk-mitigated farming decisions.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Small-Scale Farmer	I want to maximize my crop yield and income.	I Struggle to decide which crops to grow each	I want to maximize my yield	I must analyze historical agricultural data (1997–2021), I need insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk.

PS-2	Government Policy Analyst	I am designing policies to improve agricultural productivity across India	I need to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations	I am working with the Ministry of Agriculture	By using Tableau dashboards, I am trying to uncover patterns in the data to support data-driven policy decisions and allocate
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Initial Project Planning Template

Date	13 August 2026
Team ID	xxxxxx
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	4 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint -1	Data collection	USN-1	Downloading Dataset	2	High	Kuldeep	11 August 2026	17 August 2026
Sprint -1	Data collection	USN-2	Loading the dataset	1	High	Kuldeep	11 August 2026	17 August 2026
Sprint -2	Data preparation	USN-3	Clean the data	3	Low	Kuldeep	11 August 2026	17 August 2026
Sprint -1	Data visualization	USN-4	Developing visualization	5	Medium	Kuldeep	11 August 2026	17 August 2026
Sprint -1	Dashboard	USN-5	Creating Dashboard	3	High	Kuldeep	18 August 2026	24 August 2026
Sprint -2	Story	USN-6	Creating story	3	Medium	Kuldeep	18 August 2026	24 August 2026
Sprint -2	Documentation and recording	USN-7	Creating Demo	3	High	Kuldeep	18 August 2026	24 August 2026

							2026	
Sprint -2	Documentati on on and recordings	USN-8	Creating Documents	5	High	Kuldeep	18 Augu st 2026	24 August 2026

Project Initialization and Planning Phase

Date	17 August 2026
Team ID	xxxxxx
Project Title	India's Agricultural crop production Analysis
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	Visualize Trends: Identify multi decades crop output cycle, seasonal variations , and state level productivity. • Empower Stakeholders: Provide prescriptive guidance for farmers, policymakers, and agribusiness investors. Build Interactive Dashboards: Construct intuitive Tableau stories, map overlays, and drill down interfaces.
Scope	• Data Preparation: Ingestion, cleaning, and standardization of 1997–2021 agricultural data from open repositories. Geographical & Seasonal Coverage: Spans 30+ states, Union Territories, and all cropping seasons (Kharif, Rabi, Summer, Autumn, Whole Year Visual Engineering: Development of dynamic Tableau worksheets, KPI metrics, geospatialmaps,.
Problem Statement	
Description	🔍 Cleans and transforms complex multi-year agricultural datasets into standardized analytical formats. 🔍 Leverages interactive Tableau dashboards to visually track seasonal yield fluctuations and regional disparities. 🔍 Evaluates trends across diverse cropping seasons (Kharif, Rabi, Whole Year) and major Indian states.

Impact	For Farmers: Reduces financial and climatic vulnerability via validated crop-yield insights. • For Policymakers: Facilitates targeted agricultural credit, irrigation planning,.
Proposed Solution	
Approach	• Data Ingestion & Cleaning: Aggregate multi-decade raw datasets (1997–2021), clean duplicate/null records, and normalize key variables such as production quantity, cultivated area, and yield metrics. • Visual Data Architecture: Build custom dynamic calculations and hierarchical data structures within Tableau to support cross-filtering between states, districts, seasons, and years. • Dashboard Design & Deployment: Construct specialized interactive stories and drill-down visual interfaces tailored to farmers, policy analysts, and agribusiness investors.
Key Features	• Geospatial Yield Mapping: Interactive state and district maps displaying crop density, yield distribution, and regional production highlights. • Multi-Year Trend Analysis: Line and area charts tracking production volume and land usage over 24+ years to map growth trajectories. • Season-Wise Drill Downs: Comparative visual matrices isolating Kharif, Rabi, Summer, and Whole Year cropping patterns. Interactive Filtering: Cross-functional KPI cards, drop-down filters, and top N state/crop visual summaries for dynamic stakeholder exploration

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	e.g., 2 x NVIDIA V100 GPUs
Memory	RAM specifications	e.g., 8 GB
Storage	Disk space for data, models, and logs	e.g., 1 TB SSD

Software		
Frameworks	Python frameworks	e.g., Flask
Libraries	Additional libraries	e.g., scikit-learn, pandas, numpy
Development Environment	IDE, version control	e.g., Jupyter Notebook, Git
Data		
Data	Source, size, format	e.g., Kaggle dataset, 10,000 images

Data Collection and Preprocessing Phase

Date	16 August 2026
Team ID	XXXXXX
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	• Schema Inspection: Examined dataset attributes including, Season, Crop, Area (Hectares), and Production. • Dataset Profiling: Assessed missing values, structural irregularities, and value distributions across 30+ states, 600+ districts, and 50+ crop varieties from 1997 to 2021.
Data Cleaning	• Null Value Handling: Imputed missing values in Production using regional median yield rates; dropped records missing essential spatial identifiers. • Outlier & Noise Reduction: Removed corrupt entries with negative area values or impossible yield spikes caused by data entry errors
Data Transformation	Calculated the standardized productivity metric $\text{Yield (Tones/Hectare)} = \frac{\text{Production}}{\text{Area}}.$ Normalization: Grouped low-frequency minor crops into broad categories (e.g., Pulses, Oilseeds, Cereals) for macro-level trend analysis.

Data Type Conversion	• Numerical Casts: Explicitly converted Area and Production to floating-point numeric types to allow statistical aggregations. • Temporal & Categorical Casting: Formatted Crop Year into standard integer date format and converted categorical text fields into strings for dynamic visualization filtering.
Column Splitting and Merging	Standardization: Cleaned and merged inconsistent state/district naming variations resulting from geographical boundary redefinitions over time. Attribute Parsing: Split combined seasonal text attributes (e.g., combining climate tags) into standardized season labels (Kharif, Rabi, Summer, Whole Year).
Data Modeling	☐ Star Schema Architecture: Created a central Fact Tab linked to look up Dimension Tables. ☐ Relationship Building: Established primary-foreign key relationships (1: N) between dimensions and facts to optimize query speed and drill-down performance in Tableau.
Save Processed Data	☐ Storage Format: Exported clean data to optimized .csv /.parquet formats and pushed structured tables to SQL database staging tables. ☐ Tableau Extraction: Created a hyper-extracted data source file (.hyper) ready for rapid visualization render in

Data Collection and Preprocessing Phase

Date	12 August 2026
Team ID	XXXXXX
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	3 Marks

Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Source	Data Quality Issue	Severity	Resolution Plan
Dataset 01	Numerical entries are absent	Moderate	Filter dataset and impute absent numerical values.
Dataset 02	Crop-wise separated production report	High	Make a file containing all reports

Data Collection and Preprocessing Phase

Date	12 August 2026
Team ID	XXXXXX
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	2 Marks

Data Collection Plan & Raw Data Sources Identification Template

Elevate your data strategy with the Data Collection plan and the Raw Data Sources report, ensuring meticulous data curation and integrity for informed decision-making in every analysis and decision-making endeavor.

Data Collection Plan Template

Section	Description
Project Overview	The India's Agricultural Crop Production Analysis project constructs an interactive visual analytics platform to evaluate multi-decade crop output across 30+ states and 600+ districts (1997–2021). By converting raw metrics into Tableau dashboards, it provides actionable insights to help farmers, policymakers, and investors optimize yield, reduce climate risk, and direct resources effectively.
Data Collection Plan	Extract multi-year district and state agricultural records from official government open-data portals and academic repositories. Pull core spatial (State, District), temporal (Crop Year, Season), and output attributes (Area in Hectares)

Raw Data Sources Identified	Raw Data Sources Identified Ministry of Agriculture (DES / APY Corpus): Official Area, Production, and Yield statistics across 30+ states, 56+ crops, and 5 cropping seasons (1997–2021).
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Raw Data Sources Template

Source Name	Description	Location/URL	Format	Size	Access Permissions
Dataset 1	Description of the data in this source.	Link of Dataset 1	CSV	1 GB	Public
Dataset 2	Description of the data in this source.	Link of Dataset 2	Excel	1 GB	Private (with access)
Dataset 2	Official statics compiled by the directorate of economics and statics	https://desagri.gov.in/	Excel	1 GB	Public

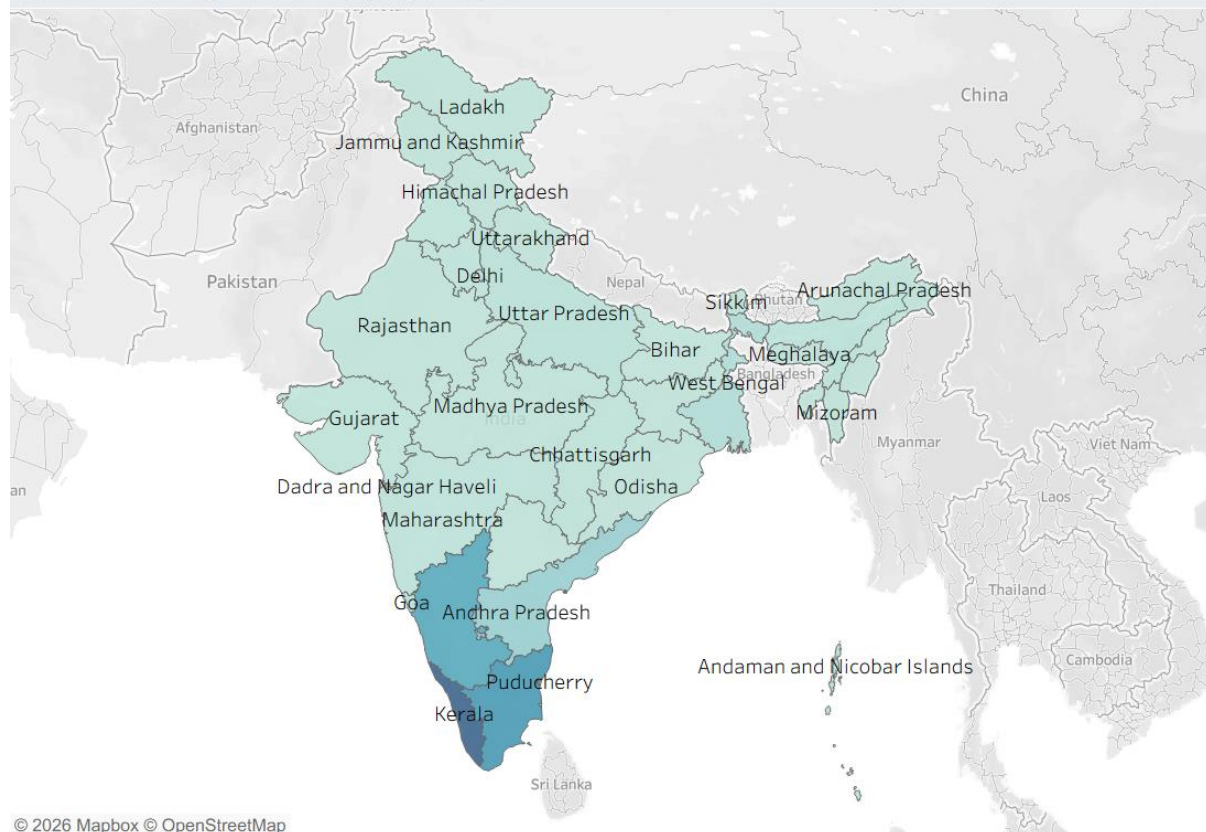
Business Question and Visualization Report

Date	14 August 2026
Team ID	
Project Name	India's Agricultural Crop Production
Maximum Marks	5 Marks

1. Which state has maximum production?

***Geographical map shows production of each state.**

state wise agriculture production



2. Which state has minimum production?

***Geographical map shows production of each state.**

state wise agriculture production



© 2026 Mapbox © OpenStreetMap

3. Which year has maximum yield?

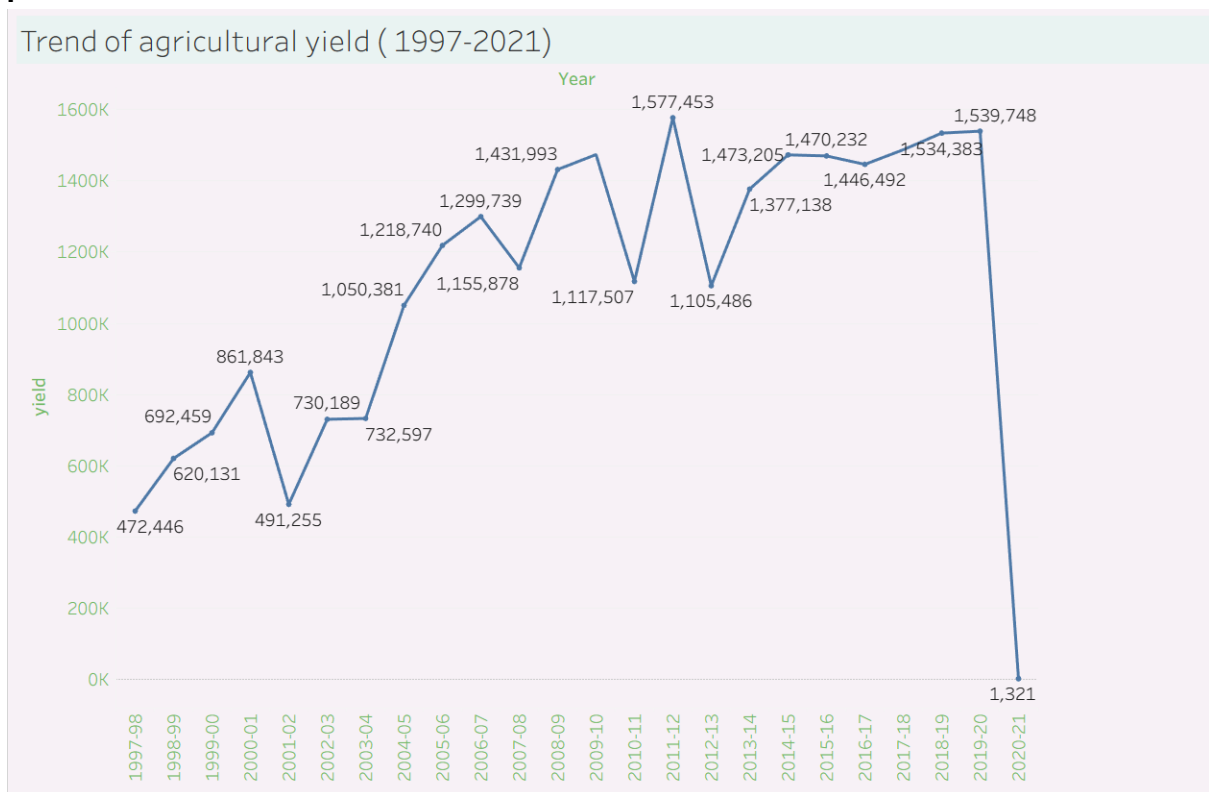
*bar chart shows year and yield

Trend of agricultural yield (1997-2021)



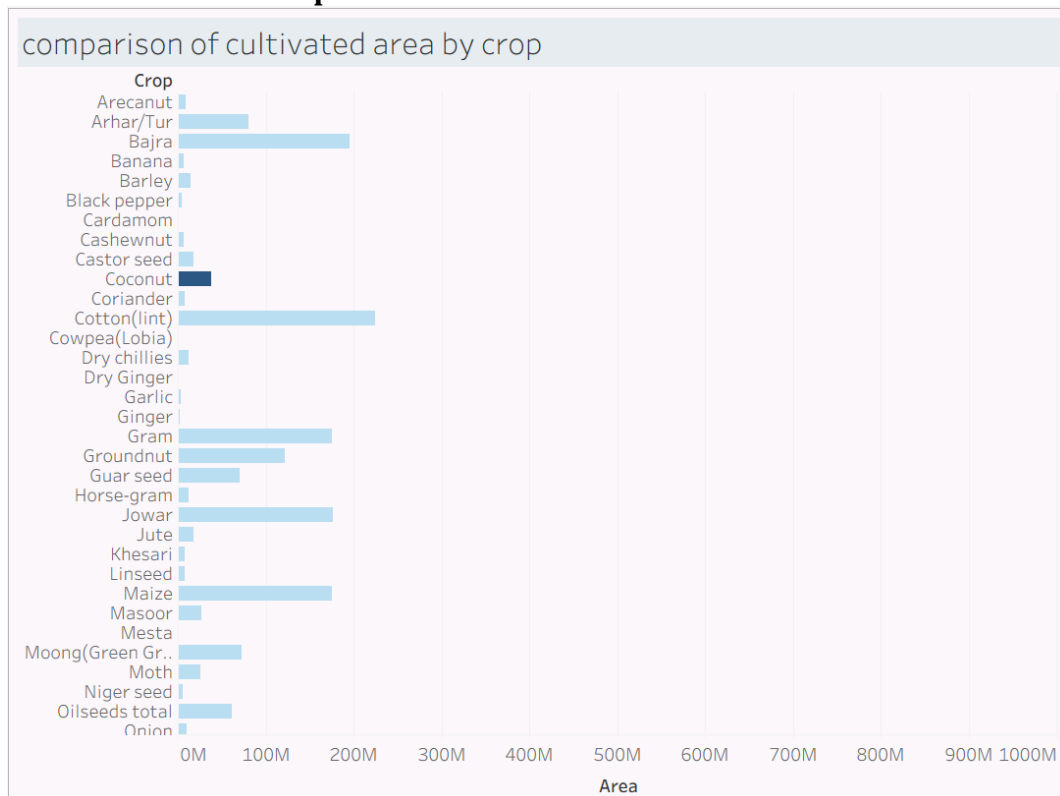
4. Which year has minimum yield?

*bar chart shows year and yield



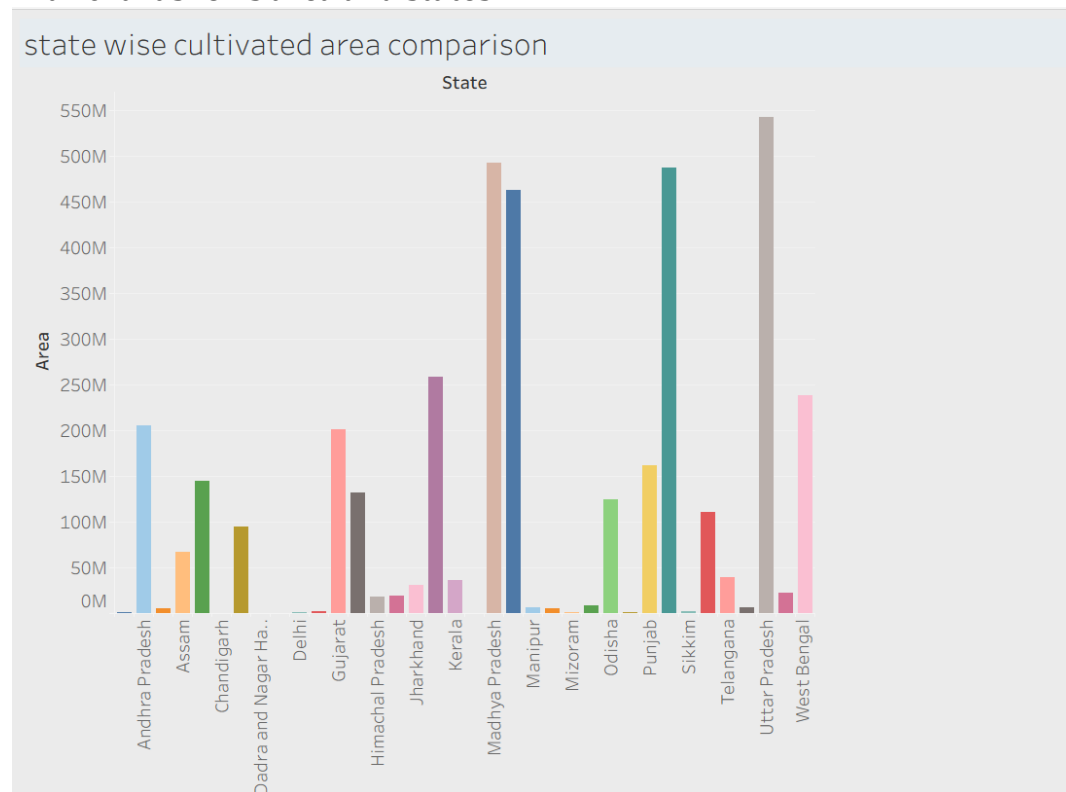
5. Which crop has maximum cultivated area?

*Bar chart shows crop and area



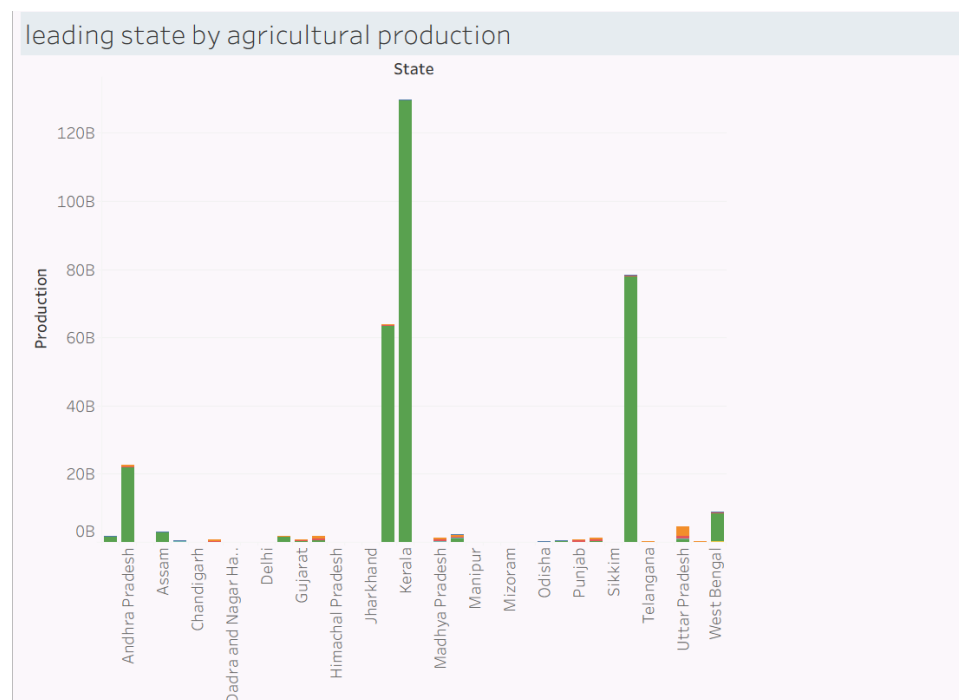
6. Which state has maximum cultivated area?

*Bar chart shows area and states



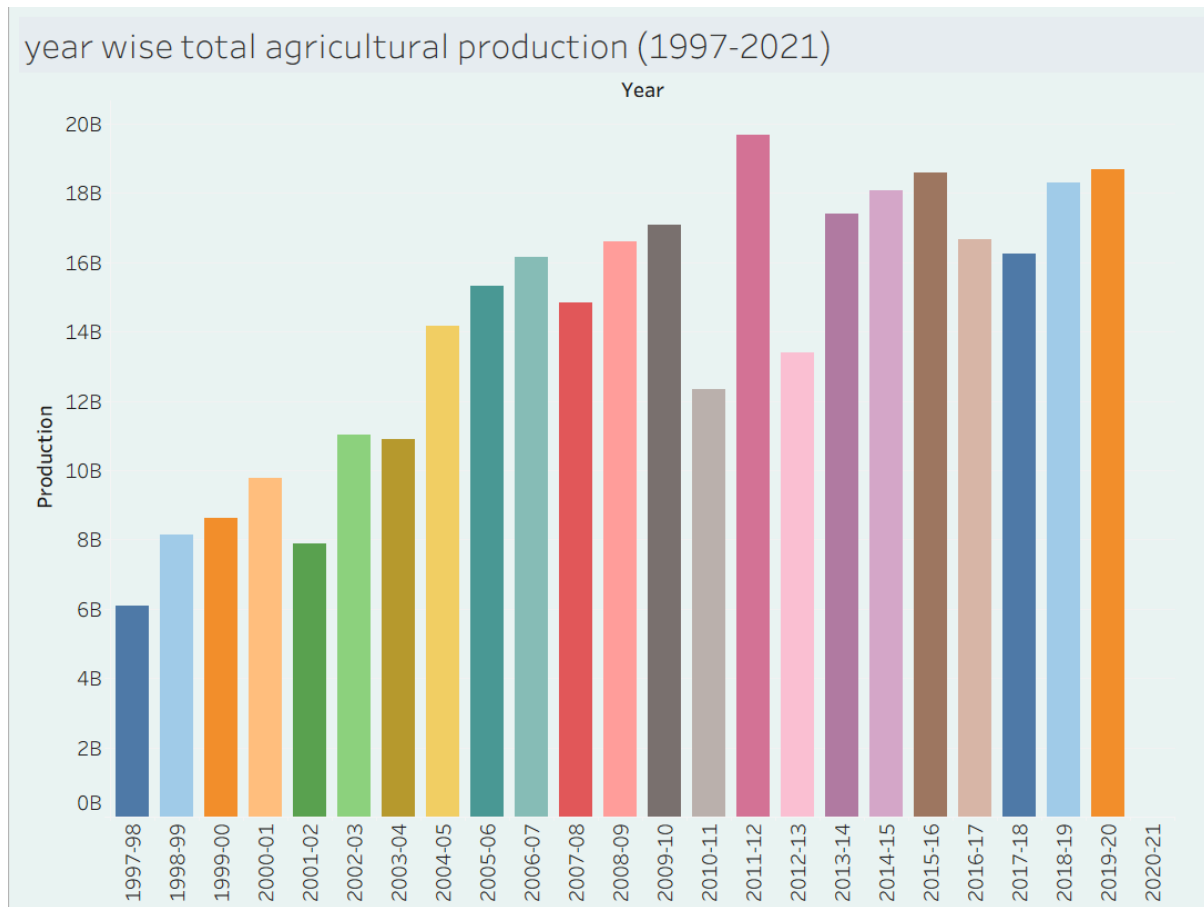
7. Which state has maximum agriculture production?

*bar chart shows leading states



8.Which year has maximum total agriculture production?

***bar chart shows year wise agriculture production.**



4.2. Developing Visualizations

1.

state wise agriculture production



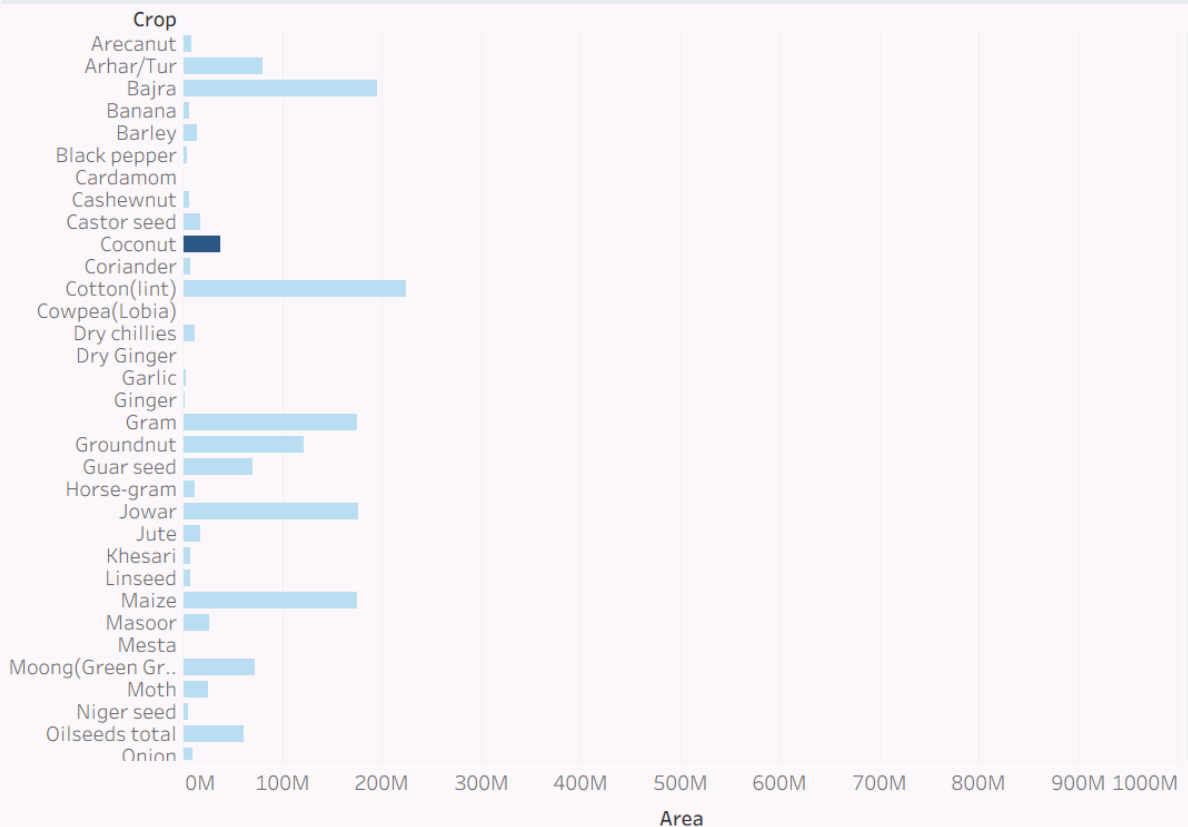
2.

Trend of agricultural yield (1997-2021)

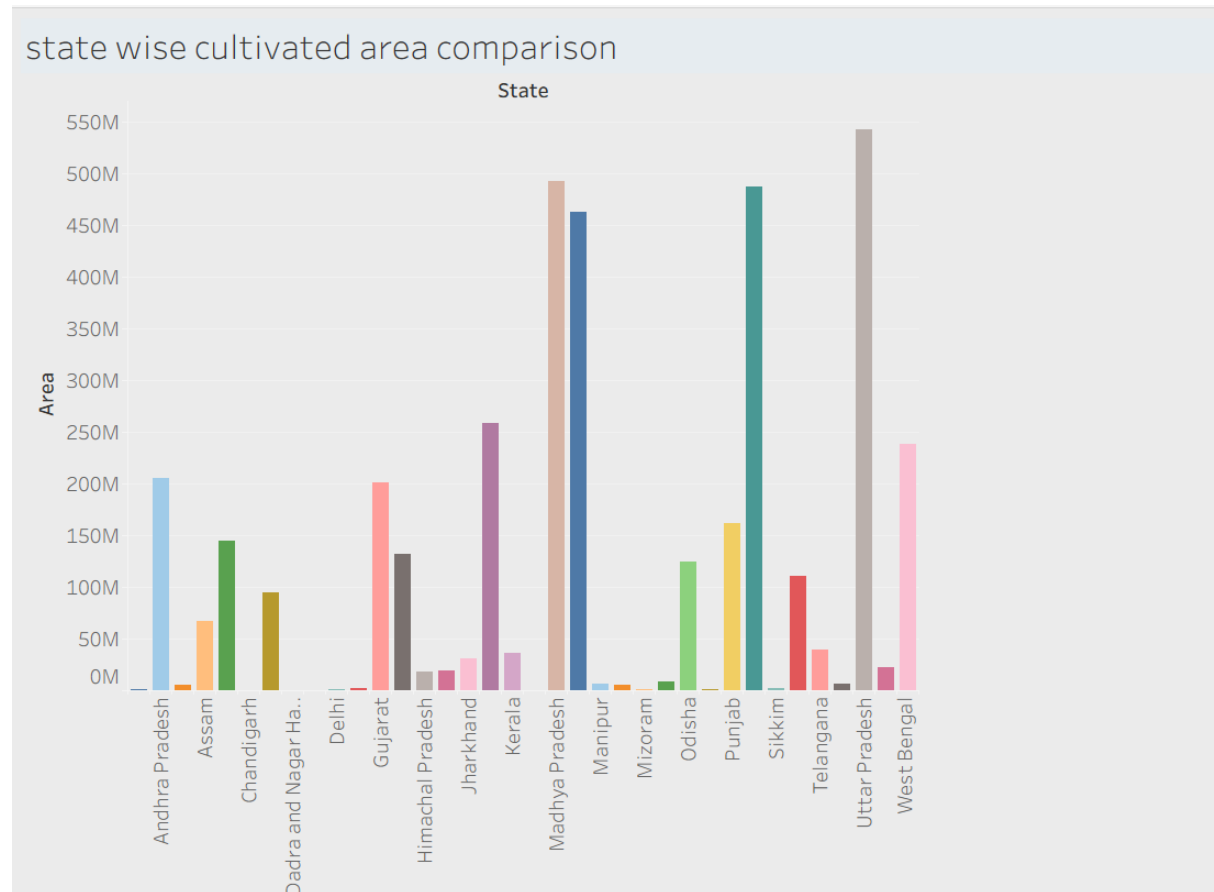


3.

comparison of cultivated area by crop

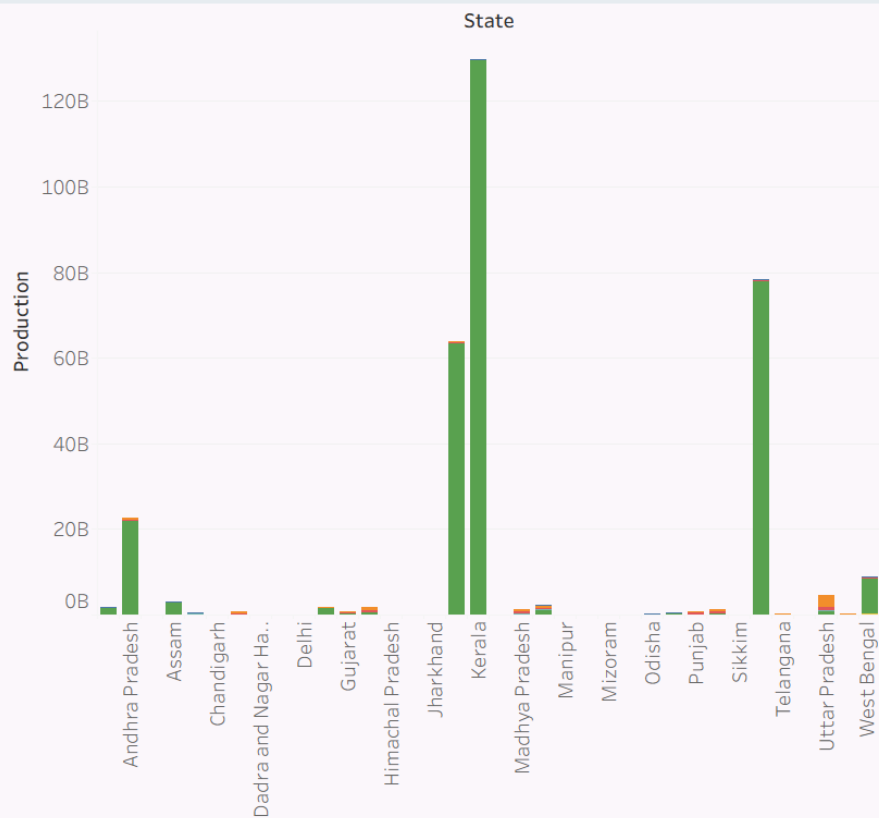


4.



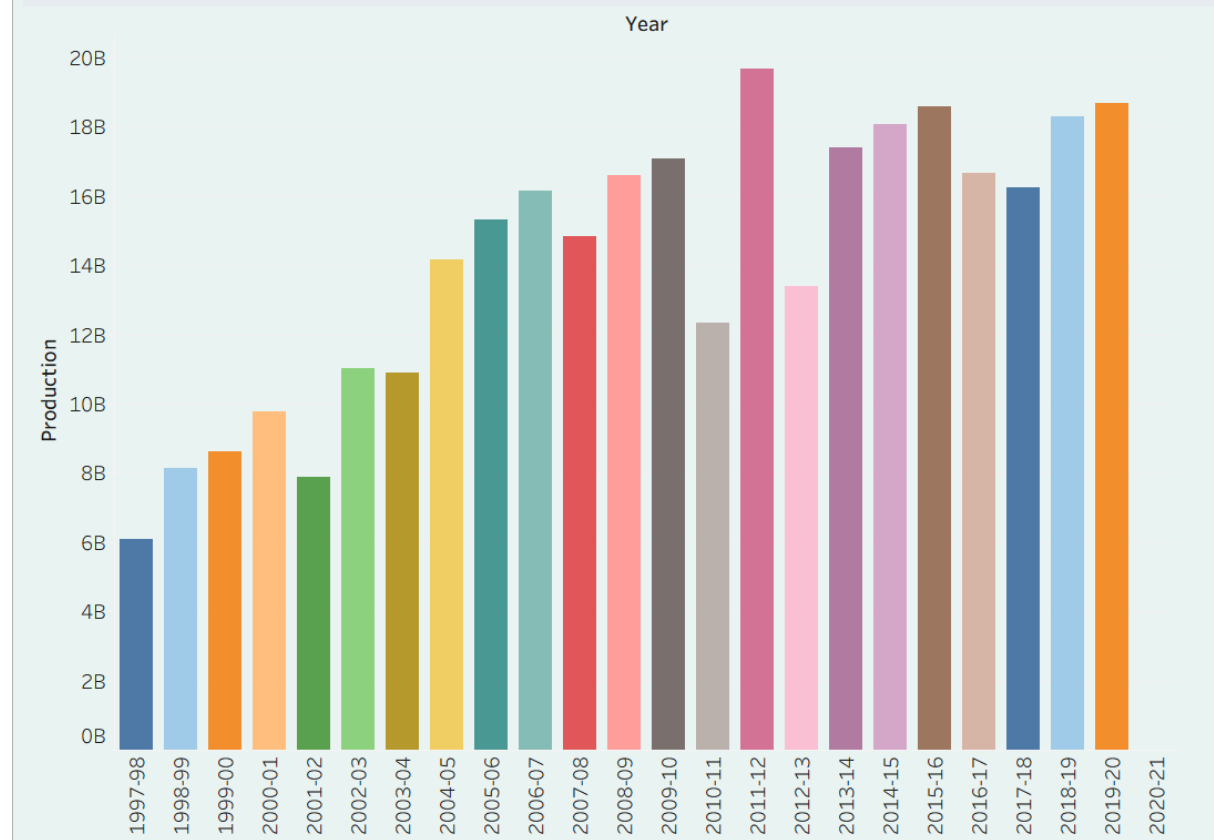
5.

leading state by agricultural production



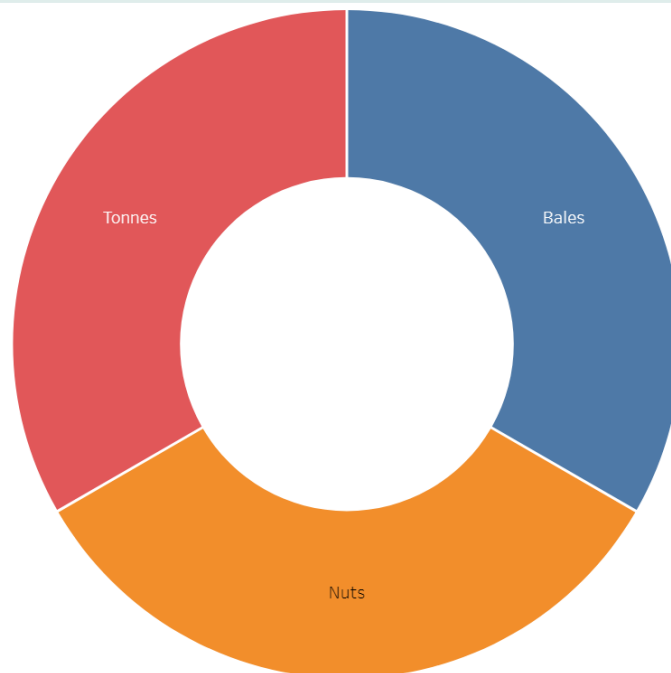
6.

year wise total agricultural production (1997-2021)



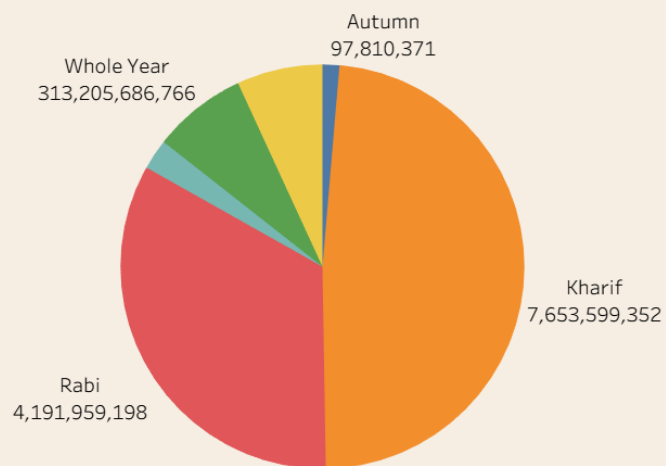
7.

production distribution by unit type



8.

season wise agricultural production share



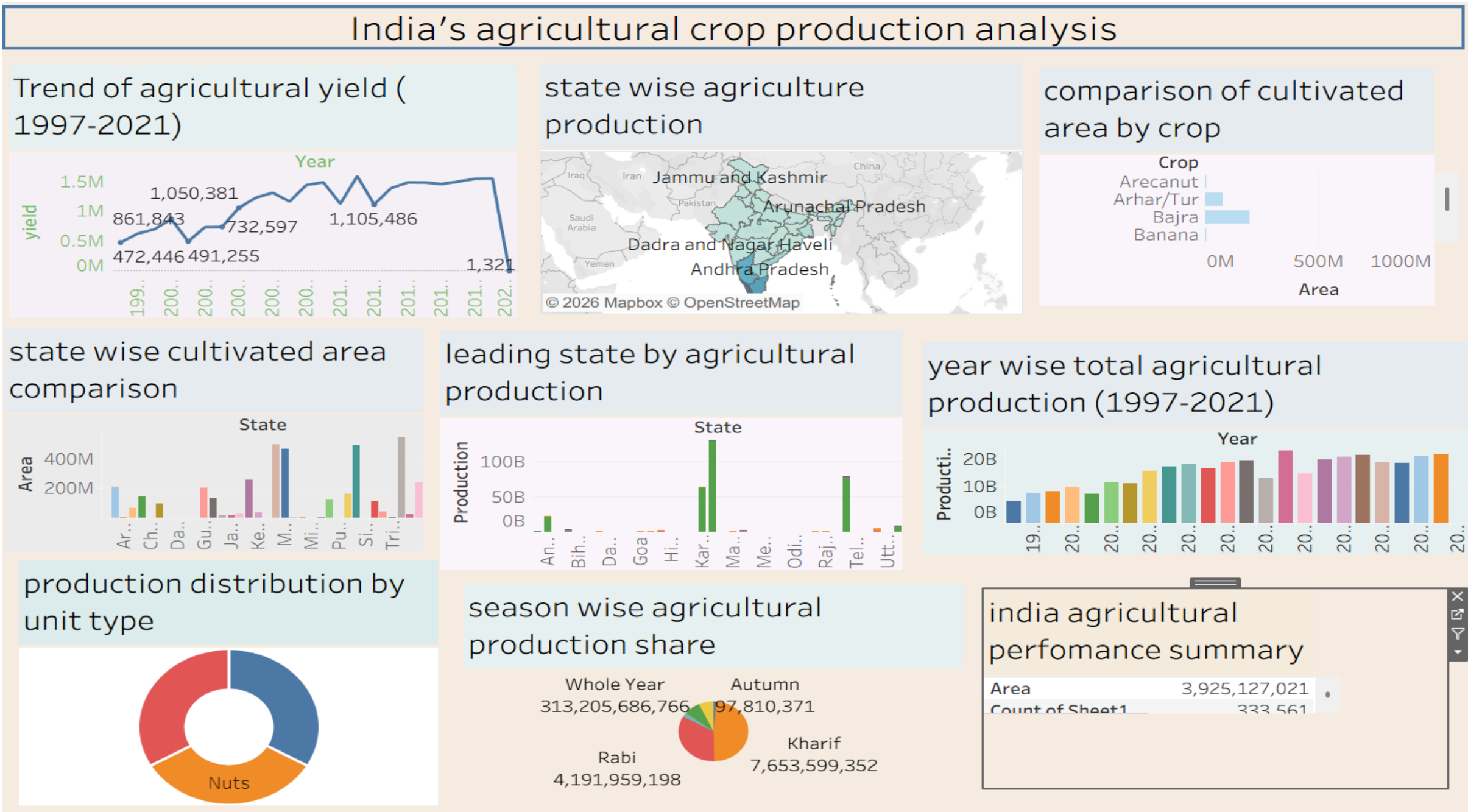
india agricultural perfomance summary

Area	3,925,127,021
Count of Sheet1	333,561
Production	326,089,923,166
yield	26,362,308

Dashboard Design

Date	20 August 2026
Team ID	
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	5 Marks

Activity 1: Interactive and visually appealing dashboards



Note: Highlight the major outcomes in form of bullet points

Sample:

Here are five potential outcomes from the dashboard image provided:

- Customer Count and Sales Overview:**The dashboard reports a total of 9,994 customers, with cumulative sales of 2.30M, a profit of 286.40K, and revenue of 11.49M. These figures highlight the overall scale and profitability of the store’s operations.
- Highest Sales:** Copiers and Machines are the top-selling products, with sales values of 2.199M and 1.646M, respectively. This suggests strong demand or high price points for these products.

- **Peak Month:** December has the highest monthly sales at 352.46K, likely due to holiday shopping.
- **Seasonal Sales Growth:** The monthly sales chart shows growth in specific months, such as March, September, October, and December, likely influenced by seasonal factors and sales events.
- **Sales Distribution Across Segments:** The **Consumer segment** leads with 0.41 M in sales, followed by Corporate at 0.25M, and Home Office at the lowest

Activity 2

Publish Dashboard on Tableau Public and Paste the Dashboard Public link below:

Example

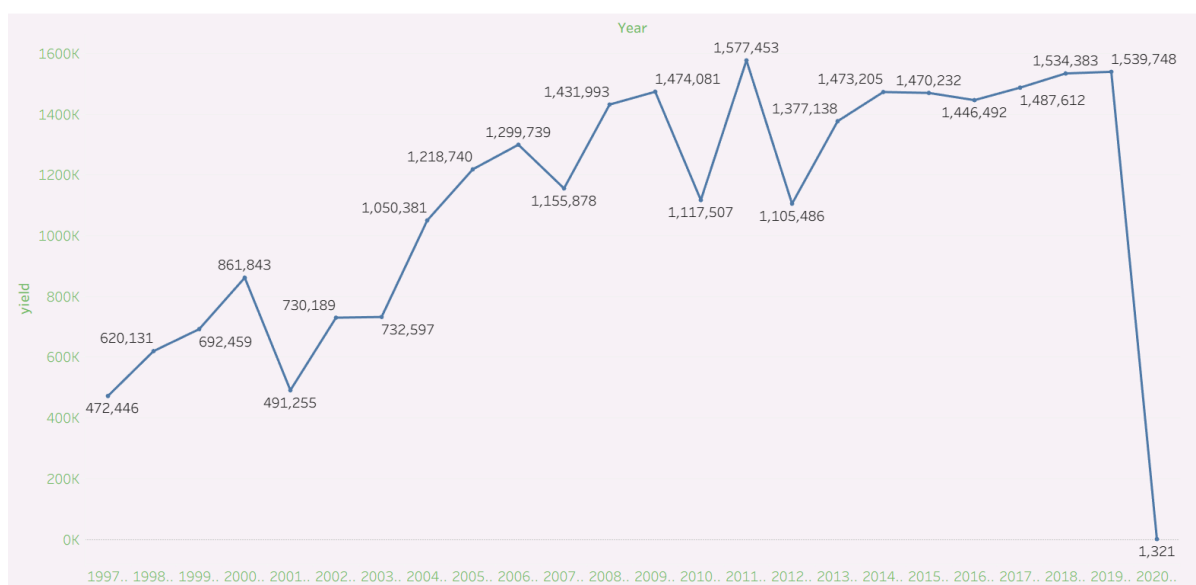
https://public.tableau.com/views/Book1_17877358352510/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Story

Date	23 August 2026
Team ID	
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	5 Marks

By using stories in Tableau, you can effectively communicate complex data in a way that is both interactive and engaging, making it easier for the audience to follow along and understand the insights. It's a tool for **data storytelling**, allowing you to present insights in a cohesive, engaging way that takes viewers through a logical progression of findings or analyses.

In Tableau, **Story** is a feature that allows you to create a sequence of dashboards, visualizations, and text to present data insights in a cohesive and narrative-driven way. It's like a slideshow within Tableau that guides the audience through a series of data points, helping them understand key insights, trends, or outcomes of your analysis.



Observations:

- **Technology Category Leads in Profit:** The Technology category is generating the highest profit among all categories.
- **Seasonal Sales Trend:** Sales peak in November, while February sees the lowest sales, indicating a possible seasonal trend.
- **Consumer Segment Dominates:** The Consumer segment contributes to the highest percentage of sales, accounting for 50.56%.
- **Top Revenue-Generating Product:** Chairs from the Furniture category generate the highest revenue, totaling 1.66M.

Activity 2:

Publish Story on Tableau Public and Paste Public link below

[https://public.tableau.com/views/Book1_17877358352510/Story1?:language=en-USPerformance Testing:](https://public.tableau.com/views/Book1_17877358352510/Story1?:language=en-USPerformance%20Testing:)

Performance Testing:

7.1 Utilization of Data filters: Data filters are used when they are required.

7.3 No of Visualization:09

8. Conclusion/Observation:

After researching on Indian agriculture, we conclude that; Indian agriculture is changing with time and agricultural yields depends upon season every year and every year the trend of seasons

is changing and we see change in seasons every year. So, we can conclude by analyzing the data that in Indian the crops should be grown in each state should be according to the environment of that state. Like Uttar Pradesh is the greatest producer of wheat so Uttar Pradesh has favorable environment for wheat production.

At the end we concluded that crops should be grown according to favorable environment and soil conditions and area of cultivation and trends of season for that particular crop in that state.

9. Future Scope: In the future, this project can be expanded to include district-level data along with state-level data. This will help in identifying areas with high and low crop production. The dashboards in future can be developed by adding crop prices and market demands. This will help us to analyze the relationship between crop production, demand, and farmers' income.

Ai will help to monitor crop diseases and faults. This can help provide early warnings so that we can prevent crops from getting damaged.

10. Appendix

10.1. Source Code (if any)

10.2. GitHub & Project Demo <https://github.com/kuldeep-200/India-s-Agricultural-production-analysis.git>